Model selection and performance estimation can diverge: a RoadSens-4M audit and a nine-acquisition PVS case study

October 26, 2026

Working manuscript, version 7 — 26 October 2026. Author names, affiliations, funding, competing-interest declarations, and a public code-deposit identifier must be supplied before submission. This draft has not been peer reviewed. RoadSens results refer to the audited Combined v3 snapshot; the PVS extension uses a separate, source-hashed release. The supplementary evidence index and reference-verification record accompany this manuscript.

Abstract

Road anomaly benchmarks can yield high within-acquisition scores while leaving performance on new acquisitions and its estimation unresolved. We audit RoadSens-4M Combined v3 using 101 CSV recordings, 616,866 source rows, two explicit label policies and four classifier families. Under the primary policy, random-forest macro-F1 changes from 0.9945 for random rows to 0.4289 for recording holdout; explicit bump-versus-pothole classification retains a contrast of 0.9948 versus 0.5189. One-second inertial statistics improve recording-heldout scores by 6.3–7.9 percentage points. Content regrouping produces mixed changes, and removing the eight specified geographic/environmental fields does not improve the tested RF recording holdouts. In a separate PVS binary speed-bump task, nested validation assesses model selection and score-estimation error on the same nine acquisitions. Relative to random-window selection, matched selection reduces primary-grid selected-system estimation MAE from 0.0530 to 0.0445 for vehicle/driver holdout and from 0.1696 to 0.1357 for scenario holdout, failing the declared two-axis criterion. Scenario-heldout macro-F1 improves from 0.7640 for random-selected systems to 0.8359 for matched-selected systems. With disjoint windows, matched-selection MAE instead worsens on both axes. Post hoc coordinate/speed ablations retain configuration-dependent signs. The evidence distinguishes candidate selection from performance estimation and supports explicit evaluation targets rather than a universal validation or context-removal rule. Five normal-only RoadSens sources, nine PVS acquisitions and no aligned external three-class test limit inference. A fixed post hoc common-budget diagnostic retained the original failed two-axis estimation criterion; matched-selection MAE was 0.0267 for vehicle holdout and 0.1724 for scenario holdout under the common-budget outer target. This sensitivity analysis does not validate a revised method. An additional Mexico/Puebla released-subset task over two documented route families selects the same RF under random and whole-trip validation; both estimates have small errors. This counterexample further limits generalization.

Keywords: road anomaly classification; smartphone sensing; dataset audit; grouped evaluation; missingness; spatial validation; reproducibility

1. Introduction

Road anomaly classification from smartphones offers a practical setting for studying how an evaluation protocol relates to its intended use. Measurements collected close together may share a road location, recording configuration, environmental metadata, and a continuous annotation segment. Predicting another row from an already observed acquisition is therefore a different task from predicting a row in a recording absent from training. Neither task is inherently illegitimate: a road reinspection service may deliberately use location, whereas a model intended for new

collection settings requires different evidence. The distinction becomes especially important when a public package combines sensor measurements, geographic information, environmental fields, and partially populated annotations.

RoadSens-4M provides a relevant public resource [1, 2]. Our study examines a specific downloadable Combined v3 snapshot rather than treating the resource name as a fully specified benchmark. A reproducible classification task additionally requires an inventory of source files, an operational interpretation of blank labels, an explicit feature list, and exact training/test membership. A nominal sample count or a random seed alone does not provide these definitions. In particular, a CSV filename is an observable grouping unit but is not automatically evidence of an independent trip, vehicle, device, or physical road.

We investigate how these choices affect measured classification performance. The study developed in four stages. First, the RoadSens audit followed an exploratory screen and was not preregistered. Second, the initial PVS nested-selection and disjoint-window protocols were frozen before any PVS classifier fit; they assess heldout domains in a separate collection and do not transfer RoadSens-trained models. Third, the coordinate/speed ablation was frozen only after v3 PVS outcomes had been inspected, so it is a post hoc diagnostic rather than an untouched confirmation. Fourth, the common-budget diagnostic was fixed after these existing outcomes had been inspected and likewise remains post hoc. Finally, a Mexico/Puebla released-subset replication used a protocol, code and exact index map frozen before fitting, following label-support checks; it supplies a further published source but only two documented route families. Our purpose is to report a bounded empirical audit, including findings that weaken a simple leakage narrative. High contextual predictability alone does not establish prohibited information use, a lower heldout score does not identify its cause, and a grouping procedure need not lower a score monotonically.

This study contributes three inspectable results. First, it defines a version-specific RoadSens classification benchmark through explicit label policies, source-field availability and content identities. Second, it quantifies how measured performance changes with evaluation scale and feature representation, using saved split memberships and diagnostic controls that retain contrary outcomes. Third, a separate PVS case study distinguishes the outer performance of a validation-selected system from the error of that system's inner performance estimate, and examines both under disjoint-window sensitivity. The accompanying prediction-level records make these comparisons reproducible. The contribution is an empirical benchmark audit using established evaluation methods.

We address three questions:

1. How do label policy, evaluation unit and feature representation change measured performance within the audited RoadSens release?
2. Which source, content, sampling and temporal controls retain the random-row/recording-heldout contrast, and which weaken a simple context-leakage explanation?
3. Within PVS's separate binary task, does deployment-matched selection improve outer-domain classification, reduce selected-system estimation error, or both, and how do these findings change with disjoint windows and post hoc coordinate/speed ablation?

The PVS case study addresses two distinct endpoints: how well a validation-selected classifier performs on a held-out domain, and how accurately its validation score estimates held-out performance. Improvement in the first endpoint does not entail improvement in the second. The declared two-axis estimation-error criterion failed in the original analysis. We retain that failure as a result and treat the later common-budget experiment as a post hoc diagnostic, not a repaired confirmatory claim.

2. Related work and scope

The RoadSens-4M data descriptor and official archive are the starting point [1, 2]. The source descriptor reports a baseline based on a selected subset of Combined files. Our reconstruction uses the complete files that can be verified in the audited package, and our exact feature and label definitions are supplied independently. We do not claim to reproduce the source paper's reported accuracy: its precise eight-file/37-feature experiment cannot be reconstructed from the evidence available in this audit, and accuracy is not interchangeable with macro-F1. Differences in cohort,

label policy, feature construction, and metric preclude subtracting the two reported scores as a measure of the source model's error.

CKDP-PINNFed is a close RoadSens-related study that addresses GPS coarsening, privacy/federated modelling, and validation beyond the original baseline [3]. It rules out broad priority claims concerning GPS processing, federated learning, or generalization assessment on RoadSens. Our narrower contribution concerns the combination of a version-specific source inventory, explicit blank-label policies, field-availability controls, and exact grouping-based prediction comparisons. The local literature-verification record distinguishes what was verified from what was not available; absence of a reported procedure in accessible material is not proof that no such experiment exists.

The broader validation literature explains why spatial, temporal, and hierarchical dependence can make an evaluation split target a different prediction problem [4, 5]. We use this principle as methodological context, not as a new discovery. Our contribution must therefore be assessed through the specificity and reproducibility of the RoadSens evidence. A further practical difference is that we examine multiple imperfect grouping units and preserve their contrary results rather than presenting one as a verified substitute for physical route independence.

STRIDE is used only for a small supplementary recording-type task [6, 7]. Its accessible labels do not supply the normal-road observations or event boundaries required for an external replication of our three-class task. It is not presented as evidence of RoadSens-to-STRIDE transfer.

PVS supplies an independent research-team collection for a different binary speed-bump task [8, 9]. Its authors already investigated unknown vehicle, driver and environment evaluation. We therefore do not claim to introduce grouped validation for PVS. Our addition compares three fixed validation selectors against outer-domain outcomes, including estimation error and a disjoint-window sensitivity. Vehicle-mounted inertial measurements and a no-speed-bump label do not constitute an aligned smartphone normal/pothole/bump benchmark.

3. Materials and methods

3.1. Audited population and sampling

The audited Combined v3 input comprises 101 complete CSV recordings: five files from the normal-only directory and 96 mixed/anomaly recordings. An aggregate file named `99.csv` is excluded from this source population. Files from the separate Isolated package are not silently pooled with Combined. The 101 files contain 616,866 original rows. We retain source rows with zero-based indices 0, 10, 20, and so forth, yielding 61,730 candidate rows. This fixed stride is a computational sampling choice; it is not a demonstration that all temporal resolutions would yield the same results. Original source row identities, labels, and the full recording time span remain available for split construction and audit.

We use *recording* to mean one CSV source file. It is an operational term throughout the manuscript. No verified trip/route/vehicle/device identifier is inferred from it. Source archives, extracted-file inventories, cache identity, and source hashes are recorded in the accompanying reproducibility inventory.

3.2. Operational label policies

The **primary policy** uses normal labels supplied by the normal-only directory and explicitly labelled pothole and bump rows from mixed recordings. Blank mixed-record rows are excluded because they do not provide confirmed normal-road truth for this analysis. The **blank-as-normal policy** instead assigns those blank mixed-record rows to Normal. It is a sensitivity assumption about the available annotations, not an additional dataset or newly validated label set.

Policy	Normal	Pothole	Bump	Total supervised rows
Primary	4,407	11,346	17,132	32,885
Blank-as-normal sensitivity	33,252	11,346	17,132	61,730

The two policies share source observations and must not be counted as independent replications. Nor can higher predictive performance determine which label policy is semantically correct. We do not extend anomaly labels using `press_duration`, assign labels to unconfirmed event boundaries, or fabricate missing geographic coordinates.

3.3. Features and preprocessing

The main comparison uses four fixed feature sets. **Full numeric** contains 42 eligible numeric fields, excluding labels, timestamps, duration-of-press fields, and source identifiers. **Context** contains eight geographic/environmental fields: latitude, longitude, vertical accuracy, GIS elevation, and four weather fields. **Instantaneous IMU** contains 12 axes from acceleration, gyroscope, gravity, and total acceleration. **Availability masks** encode whether the original value of each of the 42 fields is finite. Exact source-column names and ordering are retained in the manifests and methods-evidence file.

Eight additional random-forest ablations use latitude/longitude values (2 fields), weather (4), GIS elevation (1), full numeric minus the eight context fields (34), IMU plus latitude/longitude (14), IMU plus weather (16), latitude/longitude availability (2), and full numeric concatenated with its availability masks (84). The 34-field group is defined by this set difference; it is not asserted to remove every possible acquisition cue. Likewise, median-filled GPS values can encode missingness through their imputed constants, so the GPS-value group is not a causal isolation of location values.

Preprocessing is fitted only on the rows actually used for fitting a given fold. Numeric columns with no finite training value are removed; retained numeric columns use training medians for non-finite values. A one-column zero fallback is used if no columns remain. Availability-only features follow the original underlying-value availability rule: a mask is dropped if its corresponding numeric field has no finite fit-row observation. For the 84-field concatenated representation, the ordinary finite-column rule is applied to the concatenated matrix, which can retain an all-zero mask as a constant. Logistic regression additionally uses a training-fitted standard scaler. These conventions are held fixed, including where they make nominal feature dimensions differ from effective fold-specific dimensions.

3.4. Evaluation units and exact splits

All main evaluations use five folds. **Random-row evaluation** uses stratified row folds and can share an original recording, annotation segment, location, or overlapping source content across training and test. **Recording-heldout evaluation** uses frozen file-level assignments, with class support constraints that place one of the five normal-only sources in each test fold. All rows from a file stay together. **Time-block evaluation** divides each recording's full original time span into five equal-width bins before label filtering. Each fold tests the corresponding bin from every recording; candidate training observations in the same recording within an inclusive one-second interval of a test timestamp are removed. Training may include both earlier and later bins. This is temporal separation within known recordings, not prospective forecasting or unseen-record testing.

For a given policy, split, and seed, methods share the same eligible training rows, test rows, and capped fit rows. Unless an uncapped condition is explicitly named, training is limited to 20,000 rows by the frozen class-proportional procedure, with sampling seed `1000×model/splitseed+zero-basedfold`. Source membership, exact fit/test row arrays, and their hashes are authoritative. Seeds alone are insufficient to reconstruct the historical file partitions across software versions.

The principal comparisons use seed 42. Random-row and recording-heldout comparisons additionally use fixed seeds 43 and 44. Model randomness, splitting, and fit-row sampling change together in these checks; they are sensitivity analyses on the same observations, not independent populations.

3.5. Classifiers and capacity controls

Model	Frozen main settings
Random forest (RF100)	100 trees; maximum depth 20; minimum leaf size 5; square-root feature sampling; balanced class weights
Extra Trees	100 trees; maximum depth 20; minimum leaf size 5; square-root feature sampling; balanced class weights; no bootstrap
Histogram gradient boosting (HistGB)	learning rate 0.1; 100 iterations; at most 31 leaves; minimum leaf size 20; L2 regularization 1; balanced class weights; early stopping disabled
Logistic regression	training scaling; C=1; L-BFGS; at most 2,000 iterations; tolerance 10^{-4}; balanced class weights

Main estimators use seed 42. The forest capacity supplement uses an RF300 configuration with **300 trees, maximum depth 25, and minimum leaf size 1**. RF100 and RF300 are each evaluated with a 20,000-row cap and with all eligible training rows. RF300 changes several capacity parameters jointly and does not isolate a tree-count effect. No parameter search is performed on the reported test scores.

3.6. Source availability and content overlap

The audit rechecks source hashes, field availability, within-record constants, and cross-record content. A content edge requires an exact sequence of at least 20 consecutive retained rows with all 12 inertial axes finite. Float 64 tuples are compared after normalizing signed zero; missing values cannot create an inertial match. Labels and model scores do not define these edges. Whole-file byte identity is checked separately. The resulting graph is frozen before fitting content-component evaluations.

The graph contains three non-singleton components, with source IDs {30,31}, {45,78}, and {46,47,68,79}, and 96 components in total: five normal-only singletons and 91 mixed components. Component-heldout evaluation keeps every connected component together. It uses constrained frozen class-supported folds, with seeds 42, 43, and 44 for the specified RF full-feature checks. The primary comparison with filename holdout uses identical supervised rows and true labels. An additional 100-file cohort removes only byte-identical source 47 while preserving the other files; it is explicitly not a fully content-deduplicated cohort. Its time-block folds retain the original one-second embargo.

3.7. Explicit-anomaly and buffered spatial diagnostics

The explicit-anomaly task contains 28,478 rows from the 96 mixed recordings: 17,132 Bump and 11,346 Pothole observations with valid finite latitude and longitude. Normal-only sources and blank labels are excluded. Random folds are rebuilt for this binary population; recording folds restrict the frozen file mapping; temporal assignment uses the original source-span bins. RF100 uses the four main feature groups. This control tests whether the main contrast requires the normal-directory distinction; it is not the same three-class benchmark.

Spatial diagnostics use this same binary population, with no invalid-coordinate exclusions in the audited snapshot. Coordinates are locally projected using Earth radius 6,371,008.8 m and the cohort's mean latitude. Globally anchored square cells of widths 100 and 500 m define groups. Five-fold stratified group assignments use seed 42. For each

fold, all candidate training points within 100 m of any test point are removed using exact nearest-neighbour queries, before applying the training cap. Every resulting train/test fold supports both classes; the minimum observed fit-to-test distance exceeds 100 m. RF100 and HistGB are evaluated with full numeric, context, and instantaneous IMU features.

Unsupervised cohort coordinates define the evaluation geometry, not fitted model parameters. Different spatial cells may still belong to the same source file or content component. A buffer between observed coordinates cannot establish independence of unobserved road segments or certify genuinely new routes.

3.8. Metrics and conditional uncertainty

The principal tables report the arithmetic mean of five fold-specific macro-F1 scores. Separately, **pooled out-of-fold (OOF) macro-F1** is computed once from all heldout predictions in a configuration. If C is a confusion matrix over the task's fixed class set, macro-F1 is the mean of `2Ckk/(2Ckk+FPk+FNk)`, with zero assigned to an undefined class F 1. Applying this nonlinear function before versus after pooling generally gives different values. Balanced accuracy, class recall, confusion matrices, and source-level results accompany the main metric. Fixed three-class metrics for a one-class individual recording have a different interpretation from metrics for a fold containing all classes.

For 24 specified main-model contrasts, we use 2,000 shared bootstrap draws, seed 20261016, of the 96 frozen content components, resampling separately within the five normal-only and 91 mixed-component strata. Component confusion matrices are accumulated with each draw's multiplicities, then pooled macro-F1 differences are computed using the same weights for both predictors. The contrasts are full-feature random minus recording, full-feature random minus time-block, and recording-heldout context minus full, for four model families and two policies. Two additional RF contrasts compare recording-heldout 34-field minus full-feature predictions using the same draws.

Percentile intervals describe conditional sensitivity of predictions from fixed fitted models to this cohort's component composition. They do not refit models, remove content overlap from historical training/test splits, estimate training/CV variability, or establish a population distribution of roads or vehicles. All 2,000 draws are valid for each of the 26 reported pooled contrasts. Five normal components remain a severe constraint. We do not infer population significance by treating rows, folds, seeds, label policies, or model configurations as independent scientific samples.

3.9. Separate-source supplement and exploratory boundary

The accessible STRIDE v4 archive [6] supplies nine Bump and eight Pothole recording folders from one POCO X 2 device identifier. Each entire recording is one example. Six driving-behaviour records are excluded rather than relabelled as normal roads. For acceleration and gyroscope streams separately, we summarize vector norms by mean, standard deviation, RMS, range, and absolute maximum, yielding ten whole-record features. A separate duration-only feature is an acquisition diagnostic. No window labels or physical event boundaries are invented.

Five-fold stratified evaluation with seeds 42, 43, and 44 uses an RF100 with depth 5 and minimum leaf size 2, and scaled logistic regression. This is an offline recording-type feasibility task; it does not align RoadSens inputs or evaluate a model trained on RoadSens. Repeated splits do not turn 17 observations into 51 independent observations.

The main expansion was frozen after 330 exploratory model-fold fits. It comprises 1,580 canonical model-fold fits, including 120 RF reproduction fits already examined in the pilot. Earlier explorations also compared physical one-second windows and prefix calibration with a raw-axis one-second statistics control. A subsequent serialized-model audit identifies sklearn 1.5.2 in the historical axis-window control, whereas the current canonical environment is 1.9.0. Its scores are retained as cross-version exploratory history, not a fully controlled representation comparison. The targeted temporal extension therefore requires a same-runtime original-implementation reference. No claim is made that instantaneous IMU features exhaust the information in inertial time series.

3.10. Targeted post-draft robustness extensions

After the first manuscript and internal simulated review, we froze two additional designs. Their selection was informed by the existing results and they do not provide an untouched confirmation set. The first checks recording-heldout RF sensitivity to retaining every fifth and twentieth source row, with the original every-tenth-row analysis as the reference. Full-numeric and instantaneous-IMU features use the original source-file fold mapping, label policies, RF100 settings, training-only preprocessing, and class-proportional fit cap. Reconstructed stride-ten row identities, numeric features, and fit/test membership must match the original cache exactly before any new fits. Different strides produce different supervised and fitted row populations; their scores are compared descriptively, not as paired same-row effects.

The second benchmarks causal one-second raw-axis statistics against instantaneous IMU under recording and time-block holdout, using RF100 and HistGB with both policies. The 60 temporal statistics are computed at full source cadence before selecting the frozen stride-ten anchors. For each source and fold, we explicitly check intersections between the raw source-row sets contributing to training and test windows. Any remaining intersecting training anchor would be removed before the fit cap, and both representations use identical resulting fit rows. In the audited cohort, the existing inclusive one-second anchor embargo already yields disjoint one-second raw windows, so no additional anchors are removed. This checks source-row sharing within a recording, not equality of signal content across different files. Random-row window evaluations are excluded.

A no-fit independent check verifies window membership against a naive reference in 13 synthetic cases and 303 anchors across all 101 files. Because historical axis-window models used a different sklearn version, we first rerun the unchanged original window implementation in 1.9.0 to create a separate ten-fold reference. The first ten RF recording/window fits in the extension must reproduce this same-runtime reference's predictions and exact fit/test rows before the remaining comparisons can be interpreted. The models remain conventional fixed classifiers; this is a temporal-context diagnostic, not a comparison against all learned sequence architectures. Results and completion status of the targeted extensions are reported separately from the original 1,580-fit suite.

All targeted configurations complete and pass independent checks: 40 sampling fits, 80 temporal fits, and 10 same-runtime original-window reference fits. Combined with the original suite, this gives 1,710 canonical computational fit-folds. The temporal block includes 40 instantaneous controls reproducing A and ten RF/window cells independently reproducing the extra ten reference fits. These redundant computational checks do not increase the number of acquisition sources or independent scientific replications. Incomplete serialization and cross-version gate attempts are retained and excluded from the completion count.

3.11. PVS case study and validation-estimate assessment

The official PVS collection [8, 9] crosses three vehicle/driver pairs with three author-defined scenarios in nine acquisitions. Vehicle and driver are perfectly confounded. We use the left-network dashboard accelerometer and gyroscope, six axes sampled at a nominal 100 Hz, and the paired sample-index label tables. Source checks require equal sensor/label lengths, chronological timestamps and declared column identities; no ambiguous nearest-time label join is performed. These checks verify the released tables' structural correspondence, not independently re-annotated physical events. The task is `no_speed_bump` versus either asphalt or cobblestone speed-bump labels. No-speed-bump means the released task's non-bump interval category; it is not general normal-road ground truth, and no pothole label is constructed.

Each acquisition contributes trailing 100-row windows at anchors 99, 124, 149 and so forth. The label is the released label at the final anchor, rather than a statement that every row in the window belongs to one event. Each axis supplies mean, population standard deviation, RMS, range and maximum absolute magnitude, giving 30 sensor features. The contextual variant adds only anchor latitude, longitude and speed. Timestamps, indices and group keys are excluded from predictors. The resulting population contains 43,204 windows, including 682 speed-bump

anchors, from 1,080,905 source rows. A predeclared sensitivity retains anchors 99, 199, 299 and so forth, giving 10,804 pairwise-disjoint raw windows. Both grids come from the same nine acquisitions.

Four fixed candidates combine RF100 or histogram gradient boosting with sensor-only or sensor-plus-context inputs. RF100 uses the main study's 100-tree, depth-20, leaf-size-5, square-root-feature and balanced-class settings. HGB uses 100 iterations, 31 leaves, learning rate 0.1, L2 regularization 1, balanced classes and no early stopping. Both use seed 42 and median imputation fitted only on the actual training rows. A deterministic, proportional class cap limits each fit to 20,000 windows; test populations are not capped or balanced.

Each outer axis has three folds, leaving either one vehicle/driver pair or one scenario out, with three whole acquisitions per test fold. The six development acquisitions undergo three inner validation schemes: stratified random-window two-fold validation, leave-one-trip-out six-fold validation, and leave-one-remaining-domain-out two-fold validation matched to the outer axis. All candidates use the same exact split and cap indices. Each selector chooses the highest pooled inner OOF macro-F1, with a fixed candidate-order tie break. All choices are saved before outer predictions. Each of the four candidates is refitted once on the development population; the unselected candidates support prespecified diagnostics and an oracle-regret description, not post-test model selection. Each grid requires 264 fits, totaling 528 PVS fits and 2,238 fits before the post hoc context-ablation follow-up, including the unchanged 1,710-fit pre-PVS suite. Computational volume does not increase the nine acquisition units.

The primary improvement endpoint is mean absolute estimation error across the three outer domains of each axis: the absolute difference between a selector's chosen candidate's inner macro-F1 estimate and that same candidate's outer macro-F1. The declared criterion requires matched validation to reduce this error by at least 0.02 versus random validation on both axes. Selected-model outer F1 and regret are separate secondary endpoints; lower estimation error need not improve classification. The 100-row-grid analysis is a sensitivity and never replaces the primary grid. We also report every fixed candidate's random-minus-outer contrast. Grouped splits must have zero raw-row overlap. Random splits on the 25-row grid intentionally permit shared raw input; the 100-row grid must remove it. Inner training sizes and cap binding differ, particularly for six-fold trip validation, so contrasts do not isolate grouping as a sole causal intervention. With only three outer domains per axis, we make no population-significance claim.

3.12. Post hoc GPS and speed block ablation in PVS

After the v3 PVS outcomes had been inspected, we conducted a frozen, post hoc block ablation to distinguish predictive changes associated with adding anchor speed, adding latitude/longitude, and adding both. It used the same nine acquisitions, 100-row trailing windows, two existing window grids, six saved outer-domain splits per grid, and RF100/HGB pipelines as the parent PVS analysis. Each model input contained the 30 inertial statistics plus (i) speed alone (31 columns), (ii) latitude and longitude alone (32 columns), or (iii) both GPS coordinates and speed (33 columns); the 30-column inertial-only models were reused. Here, "GPS block" means latitude/longitude only: speed is itself GPS-derived in the released data, so the speed-bearing blocks retain GPS-derived information. Values were float32, median imputation was fit on the exact parent capped training indices, and no scaling or new split/cap generation was performed. The four fixed blocks were IMU, IMU+speed, IMU+GPS, and IMU+GPS+speed. All five prespecified contrasts were retained, including conditional contrasts (GPS given speed and speed given GPS). These retrained-system contrasts are descriptive and cannot isolate a physical or causal effect of a field. This follow-up did not alter or re-run the parent nested selectors and does not reverse the failed primary two-axis estimation-error criterion.

3.13. Common-budget diagnostic

After inspecting the original results, we fixed a common-budget diagnostic on the existing disjoint-window PVS population. No new acquisitions, candidates, hyperparameters or decision thresholds were introduced. Exact saved split and capped training memberships, rather than regenerated seeds, define all allowable observations. Within each outer domain, the minimum available class-0 count and class-1 count across all ten inner training pools and

the outer training pool set an identical per-class budget for every fit in that domain. One prespecified subsample without replacement is shared by all four candidates within a split; imputation is fit on those selected observations only. All validation and test observations remain unchanged, and grouped partitions preserve the saved acquisition memberships.

The same three selectors choose among the original four candidates using pooled inner OOF macro-F1 and the original tie order. Choices are frozen before new outer fits and predictions. Four common-budget outer refits per domain are shared by the selectors. The primary diagnostic reports estimation error against these common-budget outer systems; a separate secondary result reuses the historical full-development outer predictions for the newly selected candidate. The latter retains the inner-to-refit sample-size gap. Thus the two analyses have different training-budget targets and must not be pooled. Candidate-level and domain-level values accompany the axis means. This single-seed analysis changes effective sample size and training composition together; it measures sensitivity to a common budget and cannot estimate a causal fraction of error due only to sample size. The original failure and threshold remain unchanged.

4. Results

4.1. Main evaluation contrast across models

Table 1 shows the full-numeric results. Under the primary policy, all three tree ensembles exceed 0.98 in random-row macro-F1 and score between 0.4289 and 0.4569 under filename holdout. Logistic regression has a lower random-row score and a smaller contrast. The blank-as-normal policy changes the score levels but retains the principal tree-ensemble contrast. Time-block results lie between random-row and recording-heldout results for these full-feature configurations, without making the temporal task an external evaluation.

Table 1: Full-numeric macro-F1, five-fold arithmetic means, seed 42.

Policy	Model	Random rows	Purged time blocks	Heldout recordings
Primary	RF100	0.9945	0.8095	0.4289
Primary	Extra Trees	0.9835	0.7802	0.4299
Primary	HistGB	0.9981	0.8460	0.4569
Primary	Logistic regression	0.7519	0.7042	0.4324
Blank-as-normal	RF100	0.9314	0.5775	0.3954
Blank-as-normal	Extra Trees	0.8715	0.5340	0.3978
Blank-as-normal	HistGB	0.9350	0.6198	0.4150
Blank-as-normal	Logistic regression	0.4564	0.4109	0.3404

The same-runtime RF checks yield random-minus-recording fold-mean differences of 0.5656, 0.5387, and 0.5354 under the primary policy for seeds 42–44, and 0.5360, 0.5548, and 0.5148 under blank-as-normal. The five fold-index differences are positive in each configuration; this is a descriptive consistency check rather than five independent paired source samples. Additional Extra Trees and HistGB seeds retain high random-row scores and lower recording-heldout scores. Logistic-regression behaviour remains model- and feature-dependent.

For pooled OOF predictions, the RF primary random-minus-recording contrast is 0.5827, with conditional component-resampling 95% interval [0.4363, 0.6960]. The blank-as-normal contrast is 0.5455 [0.4881, 0.6042]. These are pooled contrasts, not intervals for the differences in Table 1. All four model families have positive conditional intervals for this full-feature random-minus-recording comparison, but their effect sizes differ.

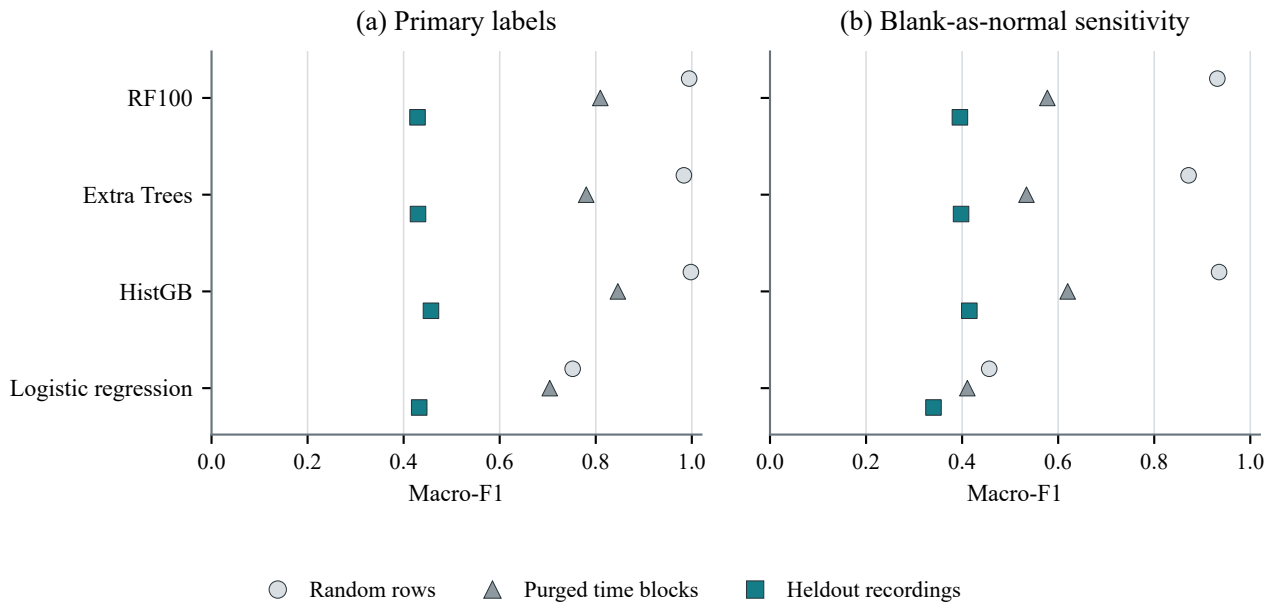


Figure 1: Full-numeric five-fold means across four model families and three evaluation scales. Points show five-fold means; they are not confidence intervals.

4.2. Label policy and feature representation

Field availability is associated with source type: the five normal-only sources lack complete GPS, while all 96 mixed sources have finite GPS. This does not imply that every Normal-labelled row lacks GPS, because blank-as-normal adds GPS-observed mixed-record rows to Normal. Four weather fields are constant within each source recording, with constants distinguished from entirely missing columns in the audit.

Availability-only RF fold-mean scores are 0.6095 for random rows and 0.6019 for recording holdout under the primary policy, but 0.2638 and 0.2720 under blank-as-normal. Thus, a strong statement that missingness alone universally solves the classification problem is unsupported. The policies change which target interpretation a source-availability pattern can predict.

Table 2 reports all eight additional RF feature groups. The GPS-value representation achieves a primary recording-heldout score of 0.6984, yet scores 0.3508 under blank-as-normal. Adding explicit masks to all numeric values yields 0.6797 and 0.3885, respectively. These are policy-dependent predictive results; the extra mask columns also change feature dimensionality and square-root sampling inside the forest.

Table 2: Eight RF ablations, five-fold mean macro-F1, seed 42.

Feature group	Primary random	Primary temporal	Primary recording	Sensitivity random	Sensitivity temporal	Sensitivity recording
GPS values	0.9985	0.8672	0.6984	0.9546	0.6180	0.3508
Weather	0.5465	0.5138	0.2958	0.4082	0.3586	0.2279
GIS elevation	0.6188	0.4354	0.3037	0.5533	0.4217	0.3888
Numeric minus context	0.9362	0.6739	0.2905	0.7792	0.4431	0.3362
IMU + GPS	0.9740	0.7953	0.7113	0.8398	0.5404	0.3929
IMU + weather	0.7814	0.6120	0.3059	0.5993	0.4155	0.3149
GPS availability	0.5233	0.5566	0.5431	0.2228	0.2310	0.2315
Numeric + availability	0.9926	0.8003	0.6797	0.9175	0.5683	0.3885

Removing the specified context fields decreases, rather than improves, the observed recording-heldout score in both policies. For pooled OOF predictions, the paired 34-field-minus-full contrasts are −0.1179 [−0.2486, −0.0208] under primary and −0.0610 [−0.1064, −0.0185] under sensitivity. These fixed-model intervals do not identify a causal effect of geographic context.

The context-minus-full contrast is also model-dependent. Under primary, its pooled conditional interval is positive for RF, negative for logistic regression, and crosses zero for Extra Trees and HistGB. Under blank-as-normal, all four corresponding intervals cross zero. Context should therefore neither be declared universally harmful nor assumed to dominate full inputs for every classifier.

4.3. Forest capacity and fit-row budget

The extra capacity/budget configurations retain high random-row scores and substantially lower recording-heldout scores (Table 3). Under primary, the RF300 capped configuration improves recording-heldout macro-F1 by 0.0485 relative to RF100 capped, but the corresponding change under sensitivity is −0.0107. Uncapping RF100 gives 0.4307 and 0.3880 for recording holdout, compared with 0.4289 and 0.3954 for its capped reference. No tested configuration makes recording holdout resemble random-row evaluation.

Table 3: Capacity/budget sensitivity, five-fold mean macro-F1. RF300 jointly changes tree count, depth, and minimum leaf size.

Policy	Forest / cap	Random	Temporal	Recording
Primary	RF100 / 20,000	0.9945	0.8095	0.4289
Primary	RF100 / uncapped	0.9964	0.8168	0.4307
Primary	RF300 / 20,000	0.9969	0.8274	0.4774
Primary	RF300 / uncapped	0.9984	0.8300	0.4502
Blank-as-normal	RF100 / 20,000	0.9314	0.5775	0.3954
Blank-as-normal	RF100 / uncapped	0.9703	0.5931	0.3880
Blank-as-normal	RF300 / 20,000	0.9572	0.5908	0.3847
Blank-as-normal	RF300 / uncapped	0.9824	0.5978	0.3788

4.4. Exact content overlap and grouped re-evaluation

Sources 46 and 47 are byte-identical. The finite-inertial audit finds eight qualifying source-pair edges within the three components described in Section 3.6. Longest exact retained-row chains include 427 rows for {30,31}, 90 for {45,78}, and 246 within {46,47,68,79}. These observations demonstrate shared numeric content without proving a

physical route or event identity. A stricter complete 42-field finite-vector audit does not find cross-record exact-row pairs; matching 12 inertial fields and matching every numeric field answer different questions.

Holding content components together does not uniformly reduce performance. Across the 12 matched-cohort comparisons of three feature groups, two models, and two policies, six pooled OOF changes are positive and six negative, ranging approximately from −0.0462 to +0.0340. Changes in fold composition and training observations accompany regrouping, so these differences are not a causal estimate of duplicate-induced optimism.

RF full-feature component-heldout pooled macro-F1 across seeds 42–44 is approximately 0.386/0.474/0.453 under primary and 0.372/0.370/0.379 under sensitivity. Removing only the byte-identical source 47 leaves a 100-file cohort with 32,838 primary and 61,484 sensitivity supervised rows. On this cohort, pooled RF full-feature random/recording/temporal scores are approximately 0.995/0.445/0.818 and 0.931/0.391/0.583, respectively. Because the cohort changes, we do not treat these as paired effects relative to the 101-file table.

4.5. Explicit-anomaly and spatial results

The binary task preserves a strong evaluation contrast without the five normal-only sources (Table 4). RF full-feature scores are 0.9948 for random rows and 0.5189 for recording holdout. The availability-only results are much lower. Accordingly, normal-directory field availability is insufficient to explain the entire random/recording contrast, although the remaining design cannot separate geographic, temporal, acquisition, and content mechanisms causally.

Table 4: Explicit Bump/Pothole classification, RF100 five-fold mean macro-F1.

Features	Random	Temporal	Recording
Full numeric	0.9948	0.7271	0.5189
Context	0.9991	0.8153	0.5559
Availability only	0.4143	0.4089	0.4029
Instantaneous IMU	0.6321	0.5107	0.5240

For this class composition, an always-majority predictor has pooled macro-F1 0.3756 and balanced accuracy 0.5. Large-sample expected macro-F1 is approximately 0.4948 for uniform random predictions and 0.5 for predictions independently drawn from the observed class frequencies. Thus, 0.5 is not a universal macro-F1 chance baseline, and a score near 0.41 should not be equated with a single generic random baseline.

Buffered spatial evaluation produces 350 cells for the 100-m grid and 119 for the 500-m grid; every retained training point is more than 100 m from observed test coordinates. Context remains predictive (Table 5). Spatial holdout is not uniformly harder than file holdout because the split units preserve different shared information.

Table 5: Explicit-anomaly buffered spatial diagnostics, five-fold mean macro-F1.

Features	Model	100-m cells	500-m cells
Full numeric	RF100	0.6349	0.6215
Full numeric	HistGB	0.6551	0.6714
Context	RF100	0.6450	0.6702
Context	HistGB	0.6806	0.7277
Instantaneous IMU	RF100	0.5150	0.4989
Instantaneous IMU	HistGB	0.5125	0.4979

These results do not support a claim that geographic features become useless after spatial separation. They also do not establish unseen-road recognition: sources and content components can cross spatial folds, and observed-point separation cannot recover missing route identifiers.

4.6. Supplementary recording types and annotated runs

On STRIDE, pooled OOF macro-F1 from the ten whole-record sensor summaries is 0.7639/0.8235/0.8235 for RF and 0.7639/0.8786/0.8211 for logistic regression across seeds 42–44. Duration-only scores are 0.5882/0.5278/0.4706 for RF and 0.3561/0.4396/0.3529 for logistic regression. These are small-sample, within-STRIDE results, not an externally validated three-class RoadSens detector. Distinct archives and the absence of byte-identical CSVs do not verify physical acquisition independence.

The RoadSens raw annotation audit identifies 149 pothole and 199 bump continuous labelled runs; all 348 runs contain retained rows. A secondary equal-run-weight summary of primary full-feature row agreement is approximately 0.952/0.247/0.597 for potholes and 0.957/0.839/0.728 for bumps under random/recording/temporal evaluation. These are summaries of agreement inside observed annotation runs, not physical event detection rates. Random training can contain other rows from the same run; no verified false-alarm-per-hour or event-localization measure follows from them.

4.7. Anchor-stride sensitivity

The sampling extension independently validates all 40 new fits, including reloaded-model predictions, exact source labels and split arrays, and serialized model hashes. Table 6 reports the complete recording-heldout RF comparison. The denser primary cohort has 65,774 supervised rows; the sparser cohort has 16,440, compared with 32,885 at stride ten. Corresponding sensitivity populations contain 123,411, 30,889, and 61,730 rows. Each cohort retains the same 101 operational sources.

Table 6: Recording-heldout RF anchor-stride sensitivity, five-fold mean macro-F1. Stride ten is the existing reference, not a newly fitted configuration.

Policy	Source-row stride	Supervised rows	Full numeric	Instantaneous IMU
Primary	5	65,774	0.4386	0.3379
Primary	10 (reference)	32,885	0.4289	0.3410
Primary	20	16,440	0.4086	0.3386
Blank-as-normal	5	123,411	0.3866	0.3432
Blank-as-normal	10 (reference)	61,730	0.3954	0.3448
Blank-as-normal	20	30,889	0.3765	0.3563

These tested densities do not produce a large improvement in recording-heldout performance. Full-feature results vary more than the IMU results under primary, and the denser policy-sensitive cohorts do not improve monotonically. The extension addresses the sampling choice for recording holdout only: it does not re-estimate a random-minus-recording contrast at each stride. Changes in row population, fit sampling, and whether the cap binds prevent interpreting these differences as a pure temporal-resolution effect. Denser anchors add no independent acquisition sources.

4.8. Causal temporal context with verified raw-input separation

All 80 temporal folds pass independent row, label, prediction, metric, preprocessing, and model-hash checks. The 40 instantaneous-IMU controls reproduce their A predictions after row-ID alignment. All ten RF/window recording folds reproduce the same-runtime original implementation, whose serialized model versions and parameters are separately verified. Table 7 presents every matched feature comparison.

Table 7: Temporal-context diagnostic, five-fold mean macro-F1. Each row compares representations with identical fit/test anchors; window input is current and past one second.

Policy	Split	Model	Instantaneous IMU12	One-second statistics60
Primary	Recording	RF100	0.3410	0.4072
Primary	Recording	HistGB	0.3379	0.4173
Primary	Time-block	RF100	0.4454	0.5929
Primary	Time-block	HistGB	0.4419	0.6015
Blank-as-normal	Recording	RF100	0.3448	0.4100
Blank-as-normal	Recording	HistGB	0.3392	0.4023
Blank-as-normal	Time-block	RF100	0.3591	0.4256
Blank-as-normal	Time-block	HistGB	0.3581	0.4294

Temporal statistics increase all eight matched fold-mean point estimates. The recording-heldout improvements range from 0.0631 to 0.0794. Pooled recording-heldout RF scores increase from 0.3423 to 0.4042 under primary and from 0.3422 to 0.4039 under sensitivity; corresponding HistGB scores increase from 0.3356 to 0.4161 and from 0.3373 to 0.3996. This supports reporting temporal information rather than interpreting weak instantaneous features as evidence that inertial signals are uninformative.

The window representation still yields recording-heldout mean macro-F1 between 0.4023 and 0.4173 in these four configurations. Time-block scores are higher, but the training/testing units and allowed shared information differ. The design neither evaluates random-row window classification nor benchmarks every learned sequence architecture. Its 60 versus 12 input dimensions also prevent attributing the improvement solely to a causal effect of time context independent of representation capacity. The verified separation concerns raw rows within a source; the cross-file content-overlap limitations remain.

4.9. PVS case study: model selection and validation-estimate accuracy

All 528 PVS fits complete and pass independent model-reload, prediction, preprocessing, split, cap and metric checks. The source population and every excluded window are indexed in the supplement. The eight axis/candidate mean random-minus-outer contrasts range from +0.0443 to +0.1696. Table 8 reports both the inner estimates and outer outcomes; these are within-PVS domain-transfer results, not RoadSens-trained model transfer.

Table 8: Primary PVS candidate performance by held-out axis

Axis	Candidate	Random CV	Trip CV	Matched CV	Outer F1	Outer BA
Scenario	HGB IMU	0.9059	0.8522	0.7817	0.8341	0.8096
Scenario	HGB IMU+context	0.9336	0.8795	0.6918	0.7640	0.7127
Scenario	RF100 IMU	0.8720	0.8241	0.7880	0.8278	0.8181
Scenario	RF100 IMU+context	0.8939	0.8442	0.6947	0.7243	0.7046
Vehicle	HGB IMU	0.9008	0.8429	0.8053	0.8351	0.8768
Vehicle	HGB IMU+context	0.9210	0.8598	0.8318	0.8679	0.8827
Vehicle	RF100 IMU	0.8680	0.8146	0.7738	0.8095	0.8579
Vehicle	RF100 IMU+context	0.8900	0.8270	0.8066	0.8352	0.8695

Table 9: Primary PVS selector estimation summary

Axis	Selector	Mean estimate	Mean outer F1	MAE	Outer-minus-inner bias
Scenario	Trip	0.8795	0.7640	0.1390	-0.1155
Scenario	Matched	0.7928	0.8359	0.1357	0.0431
Scenario	Random	0.9336	0.7640	0.1696	-0.1696
Vehicle	Trip	0.8598	0.8679	0.0304	0.0081
Vehicle	Matched	0.8318	0.8679	0.0445	0.0361
Vehicle	Random	0.9210	0.8679	0.0530	-0.0530

Note: Macro-F1 and balanced accuracy are means over three outer domains per axis. The dataset has nine trips; repeated windows and the two assessment axes are not independent replicates.

For selected systems, matched validation changes estimation MAE as follows (scenario: 0.1696 to 0.1357; vehicle: 0.0530 to 0.0445). It did not meet the declared two-axis criterion; Table 9 also reports trip validation and the actual outer scores, including cases that do not favor the matched selector. Signed bias is defined as outer minus inner macro-F1. On disjoint raw windows, the corresponding contrasts range from +0.0169 to +0.0865. The nonoverlap estimation errors are scenario: 0.1069 to 0.1290; vehicle: 0.0489 to 0.0535; matched-validation MAE is higher on both sensitivity axes. These sensitivity results use fewer windows and different fit counts per acquisition; their point estimates are not independent replications or a causal decomposition of overlap and domain shift.

As a separate selection endpoint, matched validation changes primary scenario-heldout outer macro-F1 from 0.7640 to 0.8359, while vehicle/driver-heldout macro-F1 stays at 0.8679. On disjoint windows, vehicle/driver scores change from 0.8380 to 0.8522, and scenario scores from 0.7904 to 0.8168. These descriptive selection gains do not satisfy or replace the primary estimation-error endpoint.

4.10. Post hoc PVS GPS and speed block ablation

The follow-up combines 48 newly fitted feature-block models with 48 previously verified parent predictions (96 domain-level outcomes across two grids, two axes, and two classifiers). Table 10 reports macro-F1 by feature block; Tables S17–S23 give axis means, domain-level contrasts, and vehicle-minus-scenario differences. This is an exploratory follow-up to inspected results, not an untouched confirmatory comparison. The GPS block adds latitude/longitude; speed is a distinct model input but is itself GPS-derived in the release, so a speed-bearing model does not remove all GPS-source information.

In the primary vehicle/driver-heldout analysis, both single additions improved over IMU for RF (speed +0.0209; latitude/longitude +0.0167) and HGB (speed +0.0212; latitude/longitude +0.0275); the combined block had the highest mean macro-F1 for both (RF 0.8352; HGB 0.8679).

The primary scenario-heldout pattern differs: the combined block was below both single additions for RF and HGB. Its GPS-given-speed contrast (all−speed) was -0.0898 for RF and -0.0716 for HGB; its speed-given-GPS contrast (all−GPS) was -0.0795 and -0.0462, respectively. The nonoverlap results retain counterexamples: RF's scenario-heldout GPS-given-speed contrast was +0.0147; for HGB vehicle holdout, speed and latitude/longitude alone changed macro-F1 by -0.0114 and -0.0036, while their combination changed it by +0.0202. These configuration-dependent signs argue against attributing the primary scenario pattern to GPS coordinates alone. All remaining mean and domain contrasts, including contrary signs, are retained in Tables S17–S23. This follow-up neither changes the fixed parent selections nor repairs the failed primary estimation-error endpoint.

Table 10: Macro-F1 by context feature set and outer grouping.

Grid	Axis	Model	IMU	+speed	+GPS	+GPS+speed
Primary	Vehicle	RF100	0.8095	0.8305	0.8263	0.8352
Primary	Vehicle	HGB	0.8351	0.8562	0.8625	0.8679
Primary	Scenario	RF100	0.8278	0.8141	0.8038	0.7243
Primary	Scenario	HGB	0.8341	0.8356	0.8102	0.7640
Nonoverlap	Vehicle	RF100	0.7871	0.7926	0.8022	0.8151
Nonoverlap	Vehicle	HGB	0.8282	0.8168	0.8246	0.8485
Nonoverlap	Scenario	RF100	0.7872	0.7715	0.7789	0.7862
Nonoverlap	Scenario	HGB	0.8168	0.8146	0.8074	0.7986

GPS features use the protocol-defined coordinates. Values are macro-F1 means over the three outer domains.

4.11. Post hoc common-budget diagnostic

All 264 new diagnostic fits pass independent model-reload, prediction, split, per-class-budget and training-imputation checks. The diagnostic uses the same nine acquisitions and the same four candidates. Class-specific minima range from 2,917 to 3,763 non-bump windows and 18 to 57 bump windows per fit; the two scenario domains with 18 positive windows are particularly sparse. These counts are windows, not independent physical events.

For the common-budget outer target, random-to-matched selected-system estimation MAE changes as follows (vehicle: 0.0140 to 0.0267; scenario: 0.1627 to 0.1724).

For the historical full-development outer target, random-to-matched selected-system estimation MAE changes as follows (vehicle: 0.0400 to 0.0390; scenario: 0.0941 to 0.1462).

Table 11: Post hoc PVS common-budget diagnostic: three-domain mean selected-system outcomes.

Axis	Selector	Inner F1	Outer F1	MAE
Common budget outer target				
Vehicle	Random-window	0.8581	0.8468	0.0140
Vehicle	Leave-one-trip	0.8402	0.8468	0.0217
Vehicle	Matched	0.8289	0.8526	0.0267
Scenario	Random-window	0.8699	0.7083	0.1627
Scenario	Leave-one-trip	0.8389	0.7141	0.1428
Scenario	Matched	0.7740	0.6951	0.1724
Historical full budget outer target				
Vehicle	Random-window	0.8581	0.8485	0.0400
Vehicle	Leave-one-trip	0.8402	0.8485	0.0477
Vehicle	Matched	0.8289	0.8522	0.0390
Scenario	Random-window	0.8699	0.7986	0.0941
Scenario	Leave-one-trip	0.8389	0.8064	0.0768
Scenario	Matched	0.7740	0.8168	0.1462

The two targets are reported separately in Table 11. Full domain outcomes, budgets and candidate-level errors accompany the supplement. The original two-axis criterion remains failed regardless of these diagnostic outcomes. This one-budget intervention cannot assign a causal fraction of the error to sample size, and it supplies no new independent acquisition evidence.

4.12. Additional-source released-subset replication

We executed a further, deliberately bounded evaluation on the publisher-released Mexico/Puebla subset [10, 11]. Four 9,000-position JSON files represent two documented outbound/return route families. We hold out both directions together. Route identity is documentation-derived; the JSONs contain no timestamps, GPS, dates or driver/vehicle identities, and their relationship to the paper's larger reported population remains unresolved. Thus, this is a new-source released-subset task, not a verified census of independent roads or an aligned RoadSens-trained transfer test.

Before fitting, a 50-position literal-majority window proposal failed feasibility: it retained no pothole or bump windows. The retained v1 protocol records that failure. Inspection showed source event-label runs of six positions for potholes and eleven for bumps/braking. V2 fixed ten-position, nonoverlapping windows, majority-six labels and exclusion of any braking-containing window, producing 2,504 windows for non-event/pothole/bump. Blank labels denote release non-event, not certified general normal road. Both fixed classifiers use 36 raw-acceleration summaries; vibration and identities are excluded. Within each development route, random and whole-trip two-fold selectors use equal per-class training counts. Neither supplies inner route-held-out validation: only one route is available for development in each outer fold. The supplement records all choices and exact memberships frozen before the first fit.

All 24 fits pass source, split, imputation, model-reload, prediction and numerical checks. Both selectors choose RF in both route folds; selected outer macro-F1 is 0.9934 and 0.9960 (Table 12). The selection contrast is exactly zero. Full-refit estimation MAE is 0.0019 for random versus 0.0011 for whole-trip selection; the common-budget sensitivity gives 0.0015 versus 0.0007. These small descriptive differences over two routes do not reproduce the PVS selected-model advantage or establish a universal estimation defect. They are retained as a counterexample to overgeneralizing the audit. No model/window rule was revised after scores were observed. The original PVS failure remains unchanged.

Table 12: Mexico released-subset task: full-development refit outcomes by documented held-out route.

Held-out route	Inner selector	Chosen	Inner F1	Outer F1	Abs. error
Urban (1+2)	Random	RF	0.9922	0.9934	0.0012
Urban (1+2)	Whole-trip	RF	0.9922	0.9934	0.0012
Suburban (3+4)	Random	RF	0.9934	0.9960	0.0026
Suburban (3+4)	Whole-trip	RF	0.9950	0.9960	0.0010

F1 is macro-F1 on three fixed classes (higher is better); absolute estimation error is lower-is-better. There are two documentation-derived route families, with no population confidence interval. Both selectors share each selected outer model.

5. Discussion

5.1. Match evaluation to the intended information sharing

The principal result is a difference between operational prediction problems, not a claim that every use of context is leakage. The random-row setting asks how a model predicts heldout rows when closely related acquisition information can already be observed. Recording holdout withholds a CSV-defined source. Content-component holdout adds a documented exact-content restriction. Buffered spatial evaluation withholds coordinate neighbourhoods while potentially retaining the same recording elsewhere. These designs restrict different shared relations; none is proven to recreate every deployment condition.

An application revisiting known roads may legitimately use geographic context, while a claim about new devices or routes requires verified device/route separation. Reporting a high random-row score without its information-sharing assumptions can overextend its scope. Conversely, choosing the smallest score among alternative splits is not a principled substitute for a deployment definition. The source schema, acquisition metadata, and desired use must determine the interpretation.

5.2. Label semantics and source structure should be reported together

Blank-as-normal changes the target interpretation, class composition, and relationship between normal labels and field availability. Its weaker availability-mask performance is not evidence that the alternative policy is correct; it shows how strongly a modelling conclusion can depend on that assumption. No experiment here adjudicates the physical state of the blank rows. Future benchmarks should specify whether blanks are unreviewed intervals, verified normal segments, or another annotation state and should provide provenance for any normal-road assignment.

The finite-availability and contextual controls make source associations visible. However, feature removal is a change to a predictive system, not an intervention on road condition or collection protocol. Retraining after removing columns changes forest feature sampling and interactions, and adding masks changes representation size. Our data support the observation that context removal does not improve these heldout predictions; they do not identify what would happen if the acquisition process itself were changed.

5.3. Content duplication is evidence, not a complete causal explanation

Byte identity and exact finite inertial chains are reproducible findings. They invalidate the assumption that distinct filenames alone guarantee absence of shared content. Nevertheless, our component comparisons contain both increases and decreases, and the byte-copy-removal cohort retains the main evaluation contrast. This combination motivates a source/content audit while ruling out the simpler claim that all observed degradation is caused by duplicate files. Any attribution among duplication, geographic repetition, label continuity, and recording-domain differences would require stronger controlled designs or additional acquisition metadata.

5.4. Practical benchmark reporting

A benchmark report should first state the deployment target and the information shared across train/test partitions, including whether known location is an intended input. It should then report selected-system outer-domain performance separately from the error of the validation estimate, and show how both respond to window and grouping sensitivities, including counterexamples. These reporting steps apply established evaluation methods; they do not propose a new validation algorithm.

The PVS extension makes this recommendation testable: a validation scheme should estimate the performance of the system selected by that scheme on an explicitly heldout deployment domain. Here, matched validation did not meet the declared two-axis criterion, and its nonoverlap estimation MAE is worse on both axes. Nevertheless, deployment-matched selection improves scenario-heldout outer performance under both window grids, with unchanged or higher vehicle/driver scores. Candidate ranking and accuracy of the estimated performance are distinct objectives. Trip validation also has lower primary-grid MAE than matched validation; differing training sizes and cap binding prevent attributing this to grouping alone. A small mean signed bias can coexist with large domain-level absolute errors. The PVS comparison is an empirical audit of established validation procedures. It does not validate a general evaluation-improvement algorithm. Its positive selection result and failed estimation criterion answer different questions. Even exact training-count matching cannot remove uncertainty from nine acquisitions, disentangle vehicle from driver, or establish performance on independently sampled routes. Any revised selection or estimation rule developed using these outcomes requires new untouched acquisitions for confirmation.

The PVS block ablation reinforces that coordinate and speed inputs cannot be interpreted as a single GPS switch: speed is GPS-derived in this release, and the combined-versus-single-block pattern changes across heldout axes, classifiers, and window grids. These post hoc retrained-system contrasts are descriptive; they do not identify a field-specific causal effect or support a universal coordinate-removal rule.

5.5. Limitations

First, the RoadSens audit covers one public package version, with only five normal-only sources; PVS adds a different binary task rather than an aligned three-class external test. Nominal row counts substantially exceed the number of acquisition units, and the operational file/component groups do not establish independently sampled roads. Second, the extensions were designed after an exploratory screen. Frozen configurations prevent score-based omission within the extension, but they do not create an untouched confirmatory cohort.

Third, the fixed stride, bounded model families, hand-specified hyperparameters, and operational labels restrict the conclusions. We did not optimize a universal road detector or compare all time-series networks. Existing exploratory temporal controls are retained, but this does not make the work a sequence-model benchmark. The capacity supplement changes several parameters jointly, and the spatial diagnostic is binary rather than three-class.

Fourth, the conditional bootstrap retains fitted models and historical split dependence. It does not estimate training variability, compensate for five normal components, establish causal effects, or yield independent-road significance. Neither seeds nor label policies add independent sources. Fifth, STRIDE contributes only 17 single-device recording types; there is no aligned RoadSens-to-external transfer test, verified physical-event detector, or complete, independently verified external normal/pothole/bump evaluation. The additional Mexico/Puebla subset supplies a separate non-event/pothole/bump task with near-ceiling scores and identical selected models, but only two documented route families. Missing source timing/identity metadata, fixed annotation-run durations and the unresolved release/population mapping limit its field interpretation.

PVS contributes only three vehicle/driver pairs and three scenarios, with repeated scenarios across vehicles. Vehicle and driver effects cannot be separated. The same nine acquisitions appear in the two outer-axis analyses and both window grids, so neither axes, overlapping windows nor 840 PVS fits including the context and common-budget diagnostics are independent sample-size multipliers. The study uses source annotations, a fixed small candidate set and operational deployment targets; it does not benchmark all event detectors, establish unseen-region performance,

or demonstrate a general optimum for validation fold count. The preset 0.02 threshold is a practical decision rule, not a significance test.

Finally, literature searches are finite, some source pages may be access-limited, and related work must be refreshed for the chosen venue. We do not claim universal priority, classify earlier authors' procedures as misconduct, or infer a journal decision from experimental volume.

5.6. Future evidence requirements

Additional evidence should add genuinely new acquisitions across dates, route families, vehicles and drivers, or another independently collected event-labelled dataset with authentic acquisition identifiers. Repeated passes over a road or physical anomaly must retain shared route and event identities, and a complete acquisition holdout should not be described as unseen-road evaluation unless those route and event relationships are also separated. Driver and vehicle identities should be crossed where feasible to break their current perfect confounding. Acquisition requirements should be planned using expected effects and uncertainty over the actual acquisition and route units, rather than a fixed count of windows, seeds or model fits.

A development cohort should be used to define any revised method, threshold and primary metrics. These choices and the evaluation memberships must then be frozen before an untouched cohort of whole acquisitions is assessed. Existing PVS outer results have already been inspected and cannot supply that confirmation. The additional-source task does not remove the need for new acquisitions: its two documented routes cannot establish population precision or confirm a revised algorithm. The appropriate contribution remains a bounded empirical audit with the original failed criterion reported in full.

6. Conclusion

The audited RoadSens-4M Combined v3 snapshot supports a large difference between within-recording and recording-heldout classification under explicit label definitions. This contrast persists across the tested tree ensembles, fixed seeds, explicit-anomaly controls and forest budgets. Temporal inertial summaries improve heldout scores, whereas source-field availability and exact content overlap do not by themselves explain all observed differences. Mixed regrouping and context-ablation results prevent a universal leakage or feature-removal conclusion.

Within PVS's separate binary speed-bump task, random-window estimates exceed outer-domain scores on average under both window grids. Matched selection improves scenario-heldout classification, but fails the declared two-axis estimation-error criterion; disjoint-window estimation MAE is worse on both axes. The post hoc common-budget diagnostic also leaves matched-validation MAE higher on both axes for equal-budget outer systems. It does not show that training-size differences alone explain the original failure. Coordinate/speed ablations further show configuration-dependent contextual effects. These PVS results separate selecting a useful candidate from estimating its performance accurately. The two-route Mexico/Puebla task instead gives identical selected systems and small errors for both selectors, showing that the PVS pattern is not universal. They support reporting the evaluation target, label semantics and exact acquisition membership, while leaving unseen-region performance, an aligned external three-class test and field-specific causal effects unresolved.

Data and code availability

The Mexico/Puebla publisher supplement is cited separately [10]; raw JSONs are not redistributed in this bundle. The frozen protocol, source hashes, exact window/split indices, prediction-level results and independent reconstruction checks accompany the new-source section.

RoadSens data are obtained from the official versioned Figshare resource [2] and linked author materials [1]. STRIDE is obtained from its official v4 archive [6]. PVS sensor/label sources are obtained from the official repository-linked Kaggle release [8]; raw PVS files are not included in the manuscript bundle. PVS source and derived-population hashes, exact nested splits, immutable selections and independent checks are indexed separately in `external_pvs_v1/README.md`. Local experimental manifests, source hashes, exact split identities, compact predictions, fitted preprocessing/models, and independent metric checks are indexed by `execution_inventory.json` and `reproducibilitynotes` in the accompanying workspace. The current draft does **not** have a public code DOI or repository deposit. Before submission, authors must provide an accessible code/artifact deposit, verify redistribution permissions for each artifact class, and replace this local-workspace statement with a tested public access statement. The proposed release should document how source coordinates, source identifiers, and derived predictions are handled, with applicable archive, third-party, and institutional requirements checked explicitly.

Declarations to complete before submission

Author names and contributions, affiliations, correspondence, funding, competing interests, and any institution-required ethics/data-use declaration remain to be completed by the authors. No approval number, funding source, institutional affiliation, or public-release identifier is invented in this draft.

References

[1] A. Khandakar, D. G. Michelson, S. G. Rabbani, F. B. Shafi, M. F. Ahamed, K. R. Rahman, M. A. Rahman, M. F. Nabi, M. A. Ayari, K. Khan, and P. N. Suganthan, "RoadSens-4M: A Multimodal Smartphone & Camera Dataset for Holistic Road-way Analysis," *Scientific Data*, vol. 13, Art. no. 1036, 2026. doi: 10.1038/s41597-026-07072-y. Code: `https://github.com/rabbanishaikh/One-Million-Observations-Dataset`.

[2] A. Khandakar et al., "RoadSens-4M Dataset," Figshare, version 3, 2025 [repository citation year; v3 publication-date field: 24 February 2026]. doi: 10.6084/m9.figshare.30341143.v3. CC BY 4.0.

[3] A. Khandakar, D. G. Michelson, M. M. Mia, G. M. A. Sakeef, K. Khan, and M. A. Ayari, "CKDP-PINNFed: Dual-Layer Privacy Framework Combining Contextual K-Anonymity and Differential Privacy with Physics-Informed Federated Learning for Explainable Crowdsourced Road Anomaly Detection," *Green Energy and Intelligent Transportation*, Art. no. 100433, 2026. doi: 10.1016/j.geits.2026.100433.

[4] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017. doi: 10.1111/ecog.02881.

[5] P. Ploton, F. Mortier, M. Réjou-Méchain, N. Barbier, N. Picard, V. Rossi, C. F. Dormann, G. Cornu, G. Viennois, N. Bayol, A. Lyapustin, S. Gourlet-Fleury, and R. Pélissier, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nature Communications*, vol. 11, Art. no. 4540, 2020. doi: 10.1038/s41467-020-18321-y.

[6] A. Khandakar et al., "STRIDE (Sensor Technologies for Road Insights and Driving Evaluation) Dataset," Figshare, version 4, 2024. doi: 10.6084/m9.figshare.25460755.v4. CC BY 4.0.

[7] A. Khandakar, D. G. Michelson, M. Naznine, A. Salam, M. Nahiduzzaman, K. M. Khan, P. N. Suganthan, M. A. Ayari, H. Menouar, and J. Haider, "Harnessing Smartphone Sensors for Enhanced Road Safety: A Comprehensive Dataset and Review," *Scientific Data*, vol. 12, Art. no. 418, 2025. doi: 10.1038/s41597-024-04193-0.

[8] J. Menegazzo and A. von Wangenheim, “Passive Vehicular Sensors (PVS) datasets,” official repository-linked release. https://www.kaggle.com/datasets/jefmenegazzo/pvs-passive-vehicular-sensors-datasets. Accessed 17 October 2026. CC BY-NC-ND 4.0.

[9] J. Menegazzo and A. von Wangenheim, “Speed Bump Detection Through Inertial Sensors and Deep Learning in a Multi-contextual Analysis,” *SN Computer Science*, vol. 4, Art. no. 18, 2023 [online 17 October 2022]. doi: 10.1007/s42979-022-01438-w.

[10] A. Aguilar-González and A. Medina Santiago, “CNN-Based Road Event Detection Using Multiaxial Vibration and Acceleration Signals,” *Applied Sciences*, vol.15, no.18, Art.10203, 2025. doi: 10.3390/app151810203. Publisher supplementary archive accessed 26 October 2026.

[11] A. Aguilar-González and A. Medina Santiago, “Context-Aware Road Event Detection Using Hybrid CNN–BiLSTM Networks,” *Vehicles*, vol.8, no.1, Art.4, 2026. doi: 10.3390/vehicles8010004.